Nonlinear acoustics term project(s)

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I ended up looking into two separate topics. Section 1 discusses nonlinearity in vortex beams and is potentially related to my research tasks. Section 2 discusses traffic at a stoplight and is just for fun.

1 Nonlinear vortex beam

Acoustic vortices are beams of sound possessing quantized orbital angular momentum (OAM). The OAM is quantified by the orbital number ℓ and is responsible for driving the on-axis pressure of acoustic vortices to zero. This pressure null extends across near and far fields regardless of focusing, lending acoustic vortices to the endearing moniker "doughnut beams" [1], or better, "acoustic doughnuts," adopted henceforth. Figure (1) shows the structure of the phase and magnitude of an acoustic doughnut. Acoustic

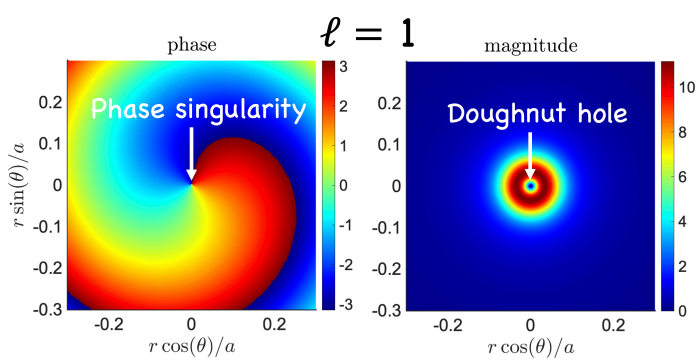


Figure 1: The phase (left) and magnitude (right) of an $\ell = 1$ acoustic doughnut at the focal plane, where the axes are normalized by the source radius a, and where r and θ are the polar coordinates. The phase singularity is at $(\frac{r}{a}\cos\theta, \frac{r}{a}\sin\theta) = (0, 0)$, which also corresponds to the location of the center of the doughnut hole, at which the pressure is 0.

doughnuts can be generated using phase plates [2], spiral gratings [3], or transducer arrays [4]. They are widely used in the context of particle manipulation [5] [6] and have more recently been considered as means to encode underwater signals [7]. Guo et al.

provide an exhaustive review on the history, theory, and applications of acoustic doughnuts [8]. *Yum!*

The present study is motivated by a recent measurement made by Richard et al. of dichromatic frequency generation near the phase singularity of an $\ell = 1$ acoustic doughnut [9]. Richard et al. attribute this dichromatic response to acoustic nonlinearity and claim to theoretically validate the measurement. Interestingly, this effect is observed at 127 dB, 67 dB lower than the nominal 194 dB threshold of nonlinearity in air. Richard et al. warn that this nonlinearity "can result in perceived Doppler interference that could lead to SONAR systems measuring that a stationary object is moving... [or] could also induce errors in time of flight measurements for any receiver with an extended aperture." A corollary of this finding is that vortex beams can potentially be used to study nonlinear phenomena at far lower amplitudes than conventionally used.

The present study investigates the feasibility of this corollary. A summary of the experiment and theory in [9] is provided in subsection 1.1. A motivating question about the results of [9] concludes subsection 1.1. Comments on the theory of Richard et al. are made in subsection 1.2, and a partially corrected equation for the field at the phase singularity given by W. Willis [10] is quoted. To answer the motivating question, the theory and notation of [9] are abandoned in subsection 1.3, in which the nonlinear acoustic doughnut is rigorously treated via numerical solution of the Westervelt equation using a MATLAB code developed by J. M. Cormack.

The $-i\omega t$ time convention is used in this work.

1.1 Summary of claim in [9]

The $\ell = 1$ acoustic doughnut discussed by Richard et al. is generated by driving the throat of a 3D-printed tapered array of waveguides (which decreases the separation between acoustic sources—see figure 3a of [9]) by a Gaussian pulse at a 40 kHz carrier frequency. At the mouth of the array of waveguides, 28 acoustic channels are independently controlled such that the outgoing phase has a helical pattern given by $\exp(i\ell\theta)$, where θ is the cylindrical polar coordinate (figure 3b). The resulting acoustic doughnut is measured 60 mm beyond the mouth of the array of waveguides (figures 3c, d), and the measured signal is provided in both time and frequency domains (figures 4b, d).

Richard et al. (at least claim to) show experimentally and theoretically that the frequency spectrum at the phase singularity of the acoustic doughnut consists of two tones, 39 kHz and 41 kHz. The fact that the input to the system is a Gaussian pulse with a 40 kHz carrier frequency implies that the generation of the acoustic doughnut is a nonlinear process [11]. To show that the nonlinear process is specifically acoustic nonlinearity,[1] Richard et al. set $\ell = 0$, for which the output is no longer an acoustic doughnut and instead consists of roughly equal-phase wavefronts. In that case, the frequency spectrum of the output is reported to match that of the input, 40 kHz, i.e., a linear response. Since the nonlinear behavior occurs for $\ell = 1$, Richard et al. conclude that there must be a

[1] in contrast to nonlinearity in the transducers, microphones, waveguides, etc.

relationship between the phase singularity and acoustic nonlinearity.

After some dubious steps, Richard et al. claim that the amplitude of the acoustic doughnut is given by

$$\boxed{\psi = \int_{r_a}^{r_b} \int_{-\pi}^{\pi} A(t) \cos[k_z z - \omega t - \Delta\phi A(t)/2\pi] r d\theta dr} \tag{1.1.1}$$

where $A(t) = P \exp\left[(t - dt)^2/\sigma^2\right]$ is the Gaussian envelope and where

$$\Delta\phi = -\omega \frac{B}{A} \frac{\Delta P}{2\rho_0 c_0^3} \frac{c_0 \Delta t}{\cos(l/kr)}$$

is the phase shift along a given path.

Box 1.1: Motivating question

Linear acoustic doughnuts have zero on-axis pressure. In fact, when an $\exp(i\ell\theta)$ phase is applied in the source plane, the on-axis pressure must necessarily be 0 in the near field.[a] Meanwhile, Richard et al. find on a *nonlinear* nonzero response at the phase singularity, which corresponds to the center of the doughnut hole, the location at which the linear pressure should vanish. This seems to imply that nonlinear acoustic behavior can occur in a region forbidden by linear acoustic behavior, namely the z axis. This is an interesting question that will be answered in subsection 1.3: *Is it indeed possible for a nonlinear response to occur on the z axis of an acoustic doughnut?*

[a] We have shown this with our analytical solution to the paraxial wave equation for a Gaussian vortex source for all z, where z is the cylindrical coordinate that corresponds to the axis of propagation. This also holds as the focal length d tends to infinity, describing an unfocused source condition. It also applies to the solution for a uniform source.

1.2 Comments on theory in [9]

Before attempting to answer the motivating question, some comments are made on the conceptual and notational issues with [9]:

(a) Willis corrects the various algebraic and notational errors of equation (1.1.1) in [10], the result of which is

$$\psi(t) = 2\pi A(t) \int_{r_a}^{r_b} \sin\left[k_z z - 2\pi f_0 t - \frac{B}{A} \frac{\pi f_0}{\rho_0 c_0^3} \frac{c_0 t}{\cos(l/kr)} A'(t)\right] r dr. \tag{1.2.1}$$

where $A'(t)$ is the surface integral of $A(t)$, i.e, $A'(t) = \int_S A(t) dS'$. Even after these extensive modifications, the result does not appear to be a solution to the Westervelt

equation, which governs the propagation of a 3D distribution of sources subject to diffraction, nonlinearity, and absorption. One cannot account for an extended source by simply modifying the phase of the source condition and integrating over area. While the surface integration over the transverse plane gives it the vague semblance of a diffraction integral, equation (1.2.1) does not account for diffraction. Nor does it account for absorption. It does account for nonlinearity through the sound speed $c = c_0 + \frac{B}{A}\frac{p}{2\rho_0 c_0}$, but since the acoustic Mach number is so small ($\epsilon \ll 1$) at the source levels used in the experiment of [9], it is implausible that the effect of nonlinearity is significant.

(b) When generalizing the acoustic doughnut to $\ell = 2$ acoustic doughnuts, Richard et al. misidentify the point of singular phase in their microphone placement. The left plot of figure (2) shows the phase of $\exp(2i\theta)$ in the focal plane of an $\ell = 2$ acoustic doughnut and denotes the true phase singularity at the point $(0, 0)$. Meanwhile, the (approximate) locations of where Richard et al. identify phase singularities are plotted as X-marks. If an $\exp(2i\theta)$ phase distribution were accurately produced, the points plotted as identified by Richards et al. as singular in phase would in fact be points of well-defined phase. The reporting of nonlinear phenomena at locations that are not in fact true phase singularities undermines the claim of [9], that nonlinear phenomena can be observed at points that truly feature singular phase.[2]

(c) Cartesian coordinates are mixed up with cylindrical polar coordinates; in particular, equation (2) of [9] has the Gaussian pulse traveling in the x direction while it should be in fact traveling in the z direction: $\psi = A(t)\sin(k_z z + \omega t + \Phi)$.

(d) Power and pressure are used interchangeably between equations (2) and (6) of [9], making it difficult to tell whether their equation (6) is a power amplitude or a pressure amplitude.

(e) As noted by [10], decibel values are provided without a reference value. Therefore, it is difficult to tell what source amplitudes are actually required to observe the report effect. Willis mentions that various other parameters are not explicitly reported by [9], making it impossible to replicate the results of Richard et al.

Another observation/question

The input pulse is propagated in a Gaussian envelope $A(t) = P\exp[(t - dt)^2/\sigma^2]$, where $\sigma = 0.5$ ms. Thus the pulse has a duration of ~ 1 ms, corresponding to a

[2] What makes this most dubious is that Richard et al. write, "Our system has relatively low spatial resolution with only 28 transducers and is mildly phase aberrated, therefore we can readily achieve spatially separated vortices by simply programming our phase array antenna to have an azimuthal phase variation of $\exp(i2\theta)$." Richard et al. utilize the poor resolution of the transducer array to construe a phase pattern that is convenient for their analysis. An $\ell = 2$ acoustic doughnut is not two $\ell = 1$ acoustic doughnuts side-by-side.

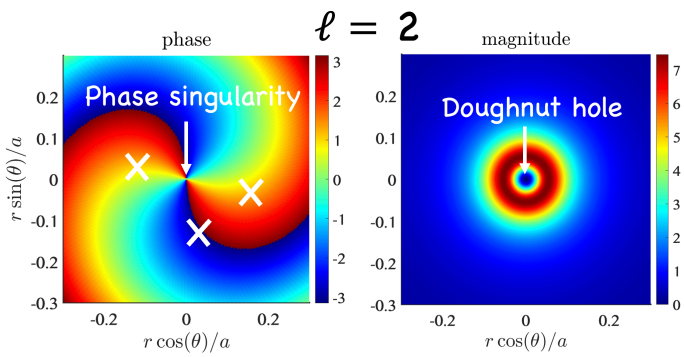


Figure 2: The phase (left) and magnitude (right) of an $\ell = 2$ acoustic doughnut at the focal plane, where the axes are normalized as in figure 1. The phase is truly only singular at the same location as in figure 1, at coordinate $(0, 0)$. The limited resolution of the transducer array used in the experiment of [9] somehow gives Richard et al. the liberty to treat the $\ell = 2$ beam as two $\ell = 1$ beams. If the $\exp(2i\theta)$ phase pattern were properly propagated through the field, the locations approximately denoted with the X-mark would in fact not be singular in phase.

> frequency of 1 kHz. Meanwhile, the generated frequencies are 1 kHz above and below the input carrier frequency, which seems to imply that the dichromatic frequency generation appears to be the result of intermodulation between the carrier frequency and the Gaussian envelope. However, in the spectrum provided in figure (4d) of 1.1, the original carrier frequency is altogether missing. This is unlike anything I have seen in nonlinear acoustics. Is it possible for the original frequency to be completely consumed by the intermodulation?

The combination of the issues listed above makes [9] very inaccessible, even after using the corrected equation (1.2.1). Willis concludes, "After making corrections to the equation but still following the same derivation, nonlinear effects had a negligible effect on simulated waveforms for the papers provided pressure values…[T]he papers results could not be reproduced" [10].

Further, the failure to account for diffraction in equation (1.2.1) makes it impossible to answer the motivating question posed in box 1.1. The following subsection abandons the theory and notation of Richard et al., instead rigorously handling the nonlinear acoustic doughnut by solving the Westervelt equation.

1.3 Rigorous treatment

To answer the question in box 1.1, a rigorous treatment of the nonlinear acoustic doughnut is now presented via numerical solution of the Westervelt equation using a MATLAB

code developed by J. M. Cormack [12].[3] While this code is a frequency-domain study and cannot comment on the intermodulation of a Gaussian pulse with a carrier frequency, it provides insight into how diffraction, absorption, and nonlinearity simultaneously affect the propagation of the acoustic doughnut. At the very least, this section should provide propitiate the reader who may be underwhelmed by the summary of and issues with [9] in subsections 1.1 and 1.2 respectively.

A solution is sought to the Westervelt equation,

$$\boxed{\frac{\partial^2 p}{\partial\tau\partial z} = \frac{c_0}{2}\nabla^2 p + \frac{\beta}{2\rho_0 c_0^3}\frac{\partial^2 p^2}{\partial\tau^2} + \frac{\delta}{2c_0^3}\frac{\partial^3 p}{\partial^3\tau},} \tag{1.3.1}$$

which accounts for diffraction, nonlinearity, and absorption to quadratic order, where τ is the retarded time $t - x/c_0$, ρ_0 is the ambient density, c_0 is the ambient sound speed, β is the coefficient of nonlinearity, and δ is the absorption coefficient [13]. The Laplacian is evaluated in Cartesian coordinates.

Note that equation (1.3.1) can be thought of as the sum of the linear wave equation in (x, y, z, τ) coordinates,

$$\frac{\partial^2 p}{\partial\tau\partial z} = \frac{c_0}{2}\nabla^2 p, \tag{1.3.2}$$

which accounts for diffraction, and the Burgers equation,

$$\frac{\partial p}{\partial z} = \frac{\delta}{2c_0^3}\frac{\partial^2 p}{\partial\tau^2} + \frac{\beta}{2\rho_0 c_0^3}\frac{\partial p^2}{\partial\tau}, \tag{1.3.3}$$

whose first term on the right-hand side accounts for absorption[4] to $\mathcal{O}(\eta\epsilon)$ and whose second term on the right-hand side accounts for nonlinearity to quadratic order. Following the algorithm outlined in [14], the solution of equation (1.3.1) is therefore found by progressing from the (τ, x, y, z) coordinate to the $(\tau, x, y, z + \Delta z)$ coordinate, first applying diffraction, then applying nonlinearity and absorption, and finally applying diffraction again. This scheme is looped to propagate the field as far in the z direction as is desired.

Diffraction is handled by assuming time-harmonic solutions to equation (1.3.2), resulting in a Helmholtz equation that is solved using Fourier acoustics [15] [16]. The field at (x, y, z) is propagated to $(x, y, z + \Delta z)$ by taking the inverse 2D spatial Fourier transform $\mathcal{F}^{-1}$ of the product

$$\hat{p}_n(k_x, k_y, z)\exp\left(i\sqrt{k_n^2 - k_x^2 - k_y^2}\Delta z - i\Delta z k_n\right), \tag{1.3.4}$$

[3] The code is written in a dimensionless form. In the explanation of the numerical solution in this section, the dimensions of the solution are retained for the sake of familiarity.

[4] $\eta = \mu\omega/\rho_0 c_0^2$ is the acoustic Stokes number

where $k_n = n\omega/c_0$, where the 2D spatial Fourier transform of $f(k_x, k_y, z)$ is

$$\hat{f}(k_x, k_y, z) = \iint_{-\infty}^{\infty} f(x, y, z) e^{-i(k_x x + k_y y)} dx dy\,,$$

and where the inverse 2D spatial Fourier transform is

$$f(x, y, z) = \frac{1}{(2\pi)^2} \iint_{-\infty}^{\infty} f(k_x, k_y, z) e^{i(k_x x + k_y y)} dk_x dk_y\,.$$

In the results presented in this work, the diffraction is handled by exactly solving the Helmholtz equation, though Cormack has included the option to use the paraxial approximation in [12] by adjusting equation (1.3.4) to

$$\hat{p}_n(k_x, k_y, z + \Delta z) = \hat{p}_n(k_x, k_y, z) \exp[-i\Delta z(k_x^2 + k_y^2)/(2k_n)]\,. \tag{1.3.5}$$

If equation (1.3.5) were used to diffract the beam, the result would be a solution to the KZK equation.

Meanwhile, nonlinearity and absorption are handled using the spectral representation Burgers equation, given by equation (10) of [14],[5]

$$\frac{\partial p_n}{\partial z} = \frac{in\beta\omega}{\rho_0 c_0^3} \left(\sum_{k=1}^{N_{\max}-n} p_k p_{n+k}^* + \frac{1}{2} \sum_{k=1}^{n-1} p_k p_{n-k} \right), \tag{1.3.6}$$

whose solution is given by the fourth-order Runge-Kutta scheme (which was used in our problem set 5 [17] except in the $+j\omega t$ time convention). The numerical code [12] was validated by comparing a previously used (lossless and linear) Fourier acoustics code to the beam patterns generated by [12] when absorption and nonlinearity were neglected for uniform circular and Gaussian source conditions.[6]

The Westervelt equation is now solved for a source condition describing that of [9]: a circular source with vortex phase shading $\exp(i\theta)$. The numerical algorithm is first set to parameters that match the experiment in [18]. These parameters are given by three dimensionless numbers, the latter two of which correspond to sound propagation underwater,

$$G = z_0/d = 10.6, \quad A = \alpha_0 d = 0.025, \quad N = d/\bar{z} = 0.212,$$

[5] We studied a dimensionless form of the spectral representation in class.

[6] For the validation, I only compared color plots of the beam amplitude vs x-z space. If we take this work further, I would modify the code to generate axial and focal line plots that compare the Fourier code to [12], to more exactly validate the code for the lossless and linear case. Interestingly, the "lumpiness" (reflections in $\boldsymbol{k}$-space?) I see with my own Fourier code appears in [12] as well.

where d is the focal length, $z_0 = ka^2/2$ is the Rayleigh distance, and $\bar{z} = \rho_0 c_0^3/\beta\omega_0 p_0$ is the shock-formation distance for a plane wave. Meanwhile, ka is set to 200, which is an average value of several vortex beam studies. The result is shown in figure 3. It is seen that the fundamental frequency recovers the beam pattern of the previously studied linear acoustic doughnut, with the on-axis null mathematically required by the source condition. Also observed is an on-axis nonzero second harmonic, which has attains its maximum $\sim$15 dB down from the fundamental. This observation seems to at least qualitatively validate the measurement of Richard et al., that it is in fact possible for nonlinear behavior to manifest along the axis of propagation, which is forbidden by the solution of the linear paraxial wave equation for a vortex source condition. A fourth harmonic also appears on-axis $\sim$35 dB down from the fundamental.[7]

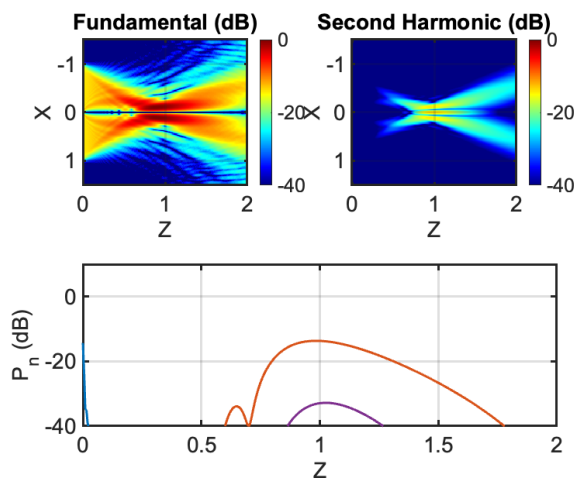


Figure 3: Solution to the Westervelt equation using [12] for $G = 10.6$, $A = 0.025$, $N = 0.212$, and $\ell = 1$. All levels are plotted as sound pressure levels [dB], and the Z-axis is nondimensionalized by the focal length. The top-left plot of the fundamental frequency shows the characteristic beam pattern of the linear acoustic doughnut, which features two maxima in the x-y plane that corresponds to the vortex ring. The bottom plot shows the axial pressure along the Z-axis, confirming the expected on-axis null for the fundamental frequency. That plot also reveals an on-axis nonzero second harmonic, which has attains its maximum $\sim$15 dB down from the fundamental at nearly the same value of Z as the vortex ring. Also, a fourth harmonic appears on-axis $\sim$35 dB down from the fundamental and attains its maximum slightly after the geometric focus.

Next, the orbital number is increased to $\ell = 2$, while G, A, N, and ka remain as before. The result is shown in figure. The beam pattern of fundamental frequency appears to spread radially outward when the orbital number is increased from $\ell = 1$ to $\ell = 2$,

[7]The color scheme for the harmonic number follows the MATLAB default outlined at `https://www.mathworks.com/help/matlab/creating_plots/specify-plot-colors.html`. Therefore, blue = 1, orange = 2, yellow = 3, purple = 4, etc.

qualitatively matching the behavior seen in an acoustic doughnut with a Gaussian focused source condition the characteristic beam pattern of the linear acoustic doughnut, for which a scaling law was derived [19]. The expected on-axis null for the fundamental frequency is again recovered. The on-axis nonzero second harmonic is also observed, appearing to have very similar behavior as in the $\ell = 1$ case. Meanwhile, the fourth harmonic that appeared in the $\ell = 1$ case is suppressed.

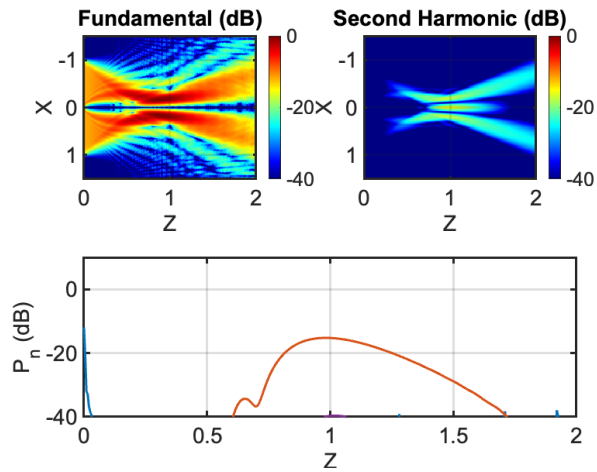


Figure 4: Solution to the Westervelt equation using [12] for $G = 10.6$, $A = 0.025$, $N = 0.212$, and $\ell = 2$. The beam pattern of the fundamental frequency appears to recover the expected linear behavior, and very little change of on-axis nonzero second harmonic is observed. Higher harmonics appear to be entirely suppressed for $\ell = 2$.

Finally, the parameters are adjusted to match the $\ell = 1$ experiment of [9]. The gain is set to 0, eliminating focusing. The drive frequency in [9] is $f = 40$ kHz, and the source size is nominally taken to be half the span of the "output waveguide phase map," (figure 3b of [9]), giving $a = 12$ mm. This gives a low $ka = 8.7$, and a Rayleigh length of $z_0 = ka^2/2 = 0.053$ m. Meanwhile, setting $\beta = 1.2$, $\rho_0 = 1.22$ kg m^{-3}, and $p_0 = 45.9$ Pa (as calculated in [10]), gives $\bar{z} = \rho_0 c_0^3/\beta\omega_0 p_0 = 3.556$ m to be the shock-formation distance for a plane wave. The absorption coefficient for 40 kHz in air is approximated to be 5 m, which may be a poor estimate. The dimensionless parameters of [18] are then normalized to the Rayleigh length:

$$A = z_0/5 = 0.0106, \quad N = z_0/\bar{z} = 0.0149.$$

These values are inputted in [12], and the result is shown in figure 3. The fundamental frequency largely recovers the previously studied the linear acoustic doughnut for $d \to \infty$. However, some reflections in $\boldsymbol{k}$-space appear to give the on-axis pressure non-zero values, most definitely due to the same numerical artifacts I have encountered previously when running the Fourier acoustics code for source distributions with nonzero orbital angular momentum. As predicted by [10], there is no nonlinear behavior to report anywhere in

the solution plane for these parameters. However, it should be noted that the nonlinearity reported by [9] is not second-harmonic generation; rather it is intermodulation between the Gaussian envelop of 1 kHz and the carrier frequency of 40 kHz. Thus the lack of a second harmonic in this numerical solution to the Westervelt equation qualitatively corroborates Willis's finding that no nonlinear behavior is observed, but does not directly refute the finding of Richard et al.

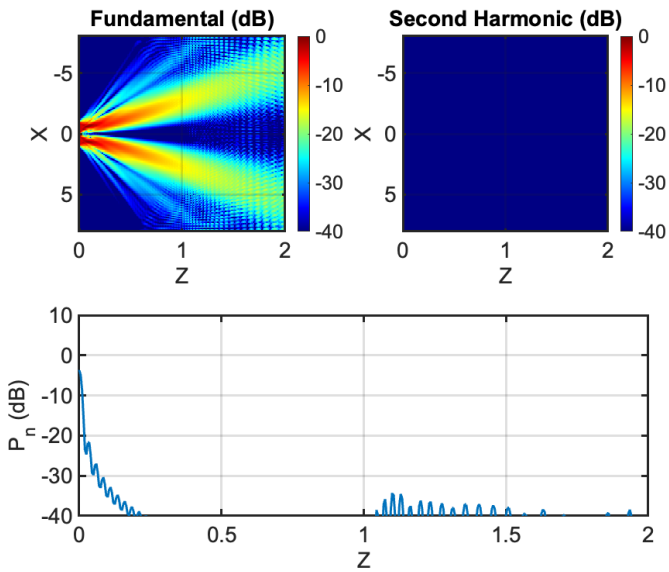


Figure 5: Solution to the Westervelt equation using [12] for $G = 0$, $A = 0.0106$, $N = 0.0149$, and $\ell = 1$. The fundamental frequency behavior mostly appears to match previous studies of the acoustic doughnut. No second harmonic is generated.

Conclusion

Unfortunately, the theory summarized in subsections 1.1 and 1.2 is not convincing, and it seems much more difficult than simply modifying the phase of a "twisting wave" to properly describe nonlinearity at the phase singularity of an acoustic doughnut. However...

Definitive answer to motivating question

A definitive answer to the motivating question posed in box 1.1 is found: "Yes, a nonlinear response *can* occur on the z axis of an acoustic doughnut, where a linear response is forbidden." This is an interesting result that at least partially corroborates the findings of [9], that the phase singularity of an acoustic doughnut can host a nonlinear response. However, as noted previously, the type of nonlinear response described by Richard et al. is the intermodulation of a carrier wave with its

envelop, not second-harmonic generation. To better assess the claim of Richard et al., a time-domain code should be used.

Applications of nonlinear acoustic doughnuts remains unclear. It seems that very little research has been performed on nonlinear doughnut beams, in both acoustics in optics. However, one short abstract does tout potential applications of second-harmonic generation of optic doughnuts [20]:

> We believe that the control of the orbital angular momentum of second harmonic beams provides an additional degree of freedom to the creative design of new nonlinear metasurfaces, with a realistic promise to increase the capacity of information encoding in compact devices for free- space communications.

J. M. Cormack's comment

One thing to note that may be interesting is that the purple curve in Fig 3 is the fourth harmonic, and not the third, suggesting that some kind of symmetry forbids odd multiples of the fundamental but allows even multiples. I would guess that as harmonics are generated off-axis and propagate towards the axis, even harmonics interfere constructively while odd harmonics interfere destructively. One could use my code to investigate, or the integral expressions in Chapter 8 if they're feeling ambitious.

Acknowledgments

Special thanks to W. Willis for his help with subsection 1.2 [10], and to J. M. Cormack for the MATLAB code used in subsection 1.3 [12].

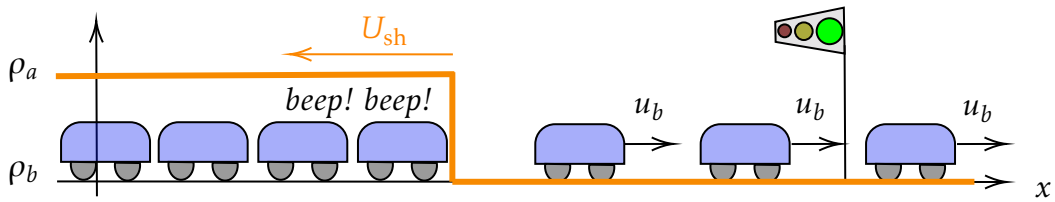
2 Traffic at a signal light

It is very frustrating to remain stationary behind a traffic signal light that has already turned green. We often blame the drivers in front of us for their inattentiveness, correlating the delayed response of the traffic flow to the green light with individual drivers' response times. Yet even if every driver were perfectly responsive, it would still take a finite amount of time for the disturbance to propagate through the line of traffic behind the light.

Motivating question

Modeling as a rarefaction shock the disturbance that passes through a line of cars when a traffic signal light turns green, is there anything we can do to speed up this shock, and hence reduce the amount of time spent stationary behind a green light?

The problem is depicted schematically below. The cars are assumed to be identical and of length l.



Subsection 2.1 re-derives the Rankine-Hugoniot relation for the conservation of mass, 2.2 presents safe-driving rules that determine the densities ahead and behind the shock, 2.3 derives the shock speed, and 2.4 provides answers to the motivating question by considering a few everyday cases.

2.1 Rankine-Hugoniot relation for conservation of mass

As discussed in class [17], the conservation of mass

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) = 0 \tag{2.1.1}$$

is of the form

$$\frac{\partial f}{\partial t} + \frac{\partial g}{\partial x} = 0. \tag{2.1.2}$$

Integrating equation (2.1.2) from x_1 to x_2 gives

$$\frac{d}{dt}\int_{x_1}^{x_2} f(x,t) + g(x_2) - g(x_1) = 0.$$

Moving the gs to the other side gives

$$I = \frac{d}{dt}\int_{x_1}^{x_2} f(x,t) = g(x_1,t) - g(x_2,t). \tag{2.1.3}$$

Let a discontinuity exist at $x_{sh}(t)$ in equation 2.1.3. The integral I is split into $I_1 + I_2$ to account for the discontinuity. The notation $x_1 < x_{sh}^- < x_{sh}^+ < x_2$ is used, where $x_{sh}^{\pm} = x_{sh} \pm \epsilon$, $\epsilon \to 0$

$$I_1 = \frac{d}{dt}\int_{x_1}^{x_{sh}^-(t)} f(x,t)dx$$
$$I_2 = \frac{d}{dt}\int_{x_{sh}^+(t)}^{x_2} f(x,t)dx.$$

Noting that for an arbitrary function $q(x,t)$,[8]

$$\frac{d}{dt}\int_{x_i}^{x^\pm} q(x,t)dx = q(x^\pm,t)\frac{dx^\pm}{dt} - q(x_i,t)\frac{dx_i}{dt} + \int_{x_i}^{x^\pm}\frac{\partial q}{\partial t}dx,$$

the integrals I_1 and I_2 are evaluated:

$$I_1 = f(x_{\text{sh}}^-,t)\frac{dx_{\text{sh}}^-}{dt} - f(x_1,t)\frac{dx_1}{dt} + \int_{x_1}^{x_{\text{sh}}^-}\frac{\partial f}{\partial t}dx$$
$$I_2 = -f(x_{\text{sh}}^+,t)\frac{dx_{\text{sh}}^+}{dt} + f(x_2,t)\frac{dx_2}{dt} - \int_{x_2}^{x_{\text{sh}}^-}\frac{\partial f}{\partial t}dx.$$

Noting that $\partial x_1(t)/\partial t = 0$ and denoting $dx_{\text{sh}}^-/dt = U_{\text{sh}}$, gives

$$I_1 = f(x_{\text{sh}}^-,t)U_{\text{sh}} + \int_{x_1}^{x_{\text{sh}}^-}\frac{\partial f}{\partial t}dx$$
$$I_2 = -f(x_{\text{sh}}^+,t)U_{\text{sh}} - \int_{x_2}^{x_{\text{sh}}^+}\frac{\partial f}{\partial t}dx$$

Taking the limit of I_1 as $x_1 \to x_{\text{sh}}^-$ gives

$$\lim_{x_1\to x_{\text{sh}}^-} I_1 = U_{\text{sh}} f(x_{\text{sh}}^-,t).$$

Similarly, taking the limit of I_2 as $x_2 \to x_{\text{sh}}^+$ gives

$$\lim_{x_2\to x_{\text{sh}}^+} I_2 = -U_{\text{sh}} f(x_{\text{sh}}^+,t).$$

Recall from equation (2.1.3) that $I = g(x_1,t) - g(x_2,t)$. Noting that $\lim_{x_1\to x_{\text{sh}}^-} I_1 = U_{\text{sh}} f(x_{\text{sh}}^-,t)$ and $\lim_{x_2\to x_{\text{sh}}^+} I_2 = -U_{\text{sh}} f(x_{\text{sh}}^+,t)$ gives

$$g(x_{\text{sh}}^-,t) - g(x_{\text{sh}}^+,t) = U_{\text{sh}}[f(x_{\text{sh}}^-,t) - f(x_{\text{sh}}^+,t)] \tag{2.1.4}$$

Equation (2.1.4) is re-written by letting the subscript a correspond to "ahead of the shock," x_{sh}^+, and by letting the subscript b correspond to "behind the shock," x_{sh}^-

$$g_b - g_a = U_{\text{sh}}(f_b - f_a),$$

or, using the jump notation $[q] = q_b - q_a$,

$$[g] = U_{\text{sh}}[f].$$

[8] Is it really called the Liebniz rule? Usually the Liebniz rule refers to the product rule.

Applying the above result to equation (2.1.1) gives the Rankine Hugoniot condition for conservation of mass:

$$\boxed{[\rho u] = U_{\text{sh}}[\rho]} \tag{2.1.5}$$

Given the density and speed ahead and behind the shock, equation (2.1.5) determines the shock speed.

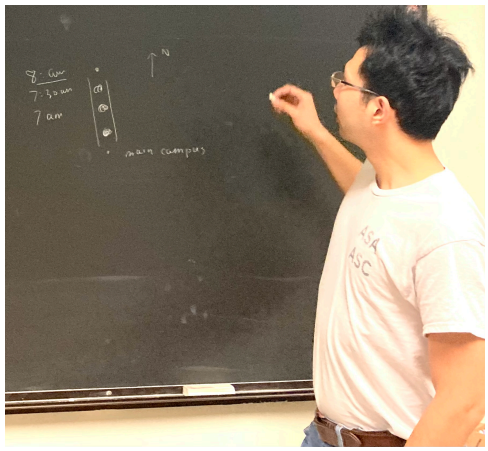


Figure 6: At the nonlinear acoustics study party, Xiaoyu explains how the shock front on Mopac southbound moves north. According to Xiaoyu, it is a noticeable effect: the later Xiaoyu leaves for campus, the further up north he meets the shock front. Thus this compression shock travels at an observable speed against the direction of traffic flow.

2.2 Safe-driving rules

The terminology "ahead" and "behind" can be confusing because the shock travels in the opposite direction as the traffic—see figure 6. In the problem at hand, ρ_a corresponds to the density of cars who have not yet experienced the shock, while ρ_b corresponds to the density of cars who have already experienced the shock.

What is ρ_a, the density of cars behind the shock? At the traffic light, suppose cars leave a distance $d \geqslant 0$ between between each other, to preserve generality. Thus the number of cars per unit length is ahead of the shock is

$$\rho_a = \frac{1}{d + l} \tag{2.2.1}$$

Meanwhile, since the cars are stationary ahead of the shock, $u_a = 0$.

What is ρ_b, the density behind of the shock? According to Dr. Hamilton, "a rule of thumb for safe driving is to stay a distance behind the car in front of you, as measured in car lengths, that is equal to your speed in mph divided by 10." This "rule-of-thumb" can be algebraically written as

$$\text{“separation distance} = l u_b/10,\text{”}$$

where l is a car length and where u_b is simply the speed limit of the road. Thus the density of cars per unit length behind the shock is therefore

$$\rho_b = \frac{1}{l(1 + u_b/10)}, \tag{2.2.2}$$

as found in homework problem 6-1.

2.3 Shock speed

The shock speed can now be determined by writing equation equation (2.1.5) as $\rho_b u_b - \rho_a u_a = U_{\text{sh}}(\rho_b - \rho_a)$ and solving for U_{sh}:

$$U_{\text{sh}} = \frac{\rho_b u_b - \rho_a u_a}{\rho_b - \rho_a}. \tag{2.3.1}$$

Inserting equations (2.2.1) and (2.2.2) into equation (2.3.1) gives

$$\boxed{U_{\text{sh}} = u_b \frac{1 + l/d}{1 - (l/d)(u_b/10)}} \tag{2.3.2}$$

Clearly, for the shock to travel infinitely fast, the cars ahead of the shock should space themselves according to the rule $\rho_a = [l(1 + u_b/10)]^{-1}$, the same rule they use when in motion at speed u_b. However, it is not practical to leave the same amount of space between cars at a stop light as when the cars are in motion. Let us conclude with a few example cases, which answer the motivating question in practical ways.

2.4 A few example cases

- The everyday scenario: people keep half car length of space in between cars on a road with a 40 mph speed limit. This gives $U_{\text{sh}} = -17$ mph. *Slow poke!* (referring to the shock)—but this is exactly what makes city traffic so frustrating. *It isn't peoples' responsiveness to the light turning green; it's the way they space their cars.* Leaving four cars' worth of space between each car would give an infinite shock speed but would lead to honking...Instead, if we compromise at two car lengths between cars, $U_{\text{sh}} = -60$ mph, which makes a good chase for the police.

- In much of north Austin, roads have higher speed limits, like ~50 mph. In this case, if we leave two car lengths between cars, the shock wave obeys the speed limit of the road, but travels in the opposite direction. Hey, I did that once!

- So if you are stopped in a school zone, it is conceivable to leave two car lengths between you and the next car, which would give $U_{\text{sh}} = -\infty$. Think about it: by spacing your car at a very safe distance of two car lengths, you're making a wave go infinitely fast, all while you're looking out for children crossing the road. The police would be ok with that, but Einstein wouldn't!

Conclusion

I had a much more sophisticated model than what is presented in section 2. It accounted for visibility of the traffic signal light due to car geometry, but this leads to a variable density ahead of the shock. Therefore the rarefaction shock is no longer a simple wave, and the Rankine-Hugoniot relations in terms of "ahead" and "behind" a step shock lose their meaning. I wanted to develop this portion of the project further, but I need to start studying for the final exam. Nevertheless, it was fun to think a bit more about the traffic problem.

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